

Hussein A. HOUDROUGE

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RESEARCH INTERESTS

Broadly, I am interested in Theoretical Computer Science and Discrete Mathematics. My current research is focused on Extremal and Structural Graph Theory. I work on problems related to graph colorings, graph parameters, and product structure theorems.

EDUCATION

Sep. 2021 - Aug. 2026	Carleton University, Ottawa, Canada Ph.D. in Computer Science, supervised by Prof. Vida Dujmovic and Prof. Pat Morin.
Sep. 2017 - Sep. 2019	Paris-Saclay University - L'Ecole Polytechnique de Paris, Palaiseau, France Master M2: Parisian Master of Research in Computer Science (MPRI) Master M1: Algorithms and Foundations of Programming languages Awarded with honor.
Sep. 2013 - Sep. 2017	The American University Of Beirut, Beirut, Lebanon Bachelor of Science in COMPUTER SCIENCE Minor in Computational Science Awarded with distinction.

PUBLICATIONS

Preprint/In Preparation:

- **3-coloring for Graphs Embedded on Surfaces**
Joint work with Vida Dujmovic and Pat Morin.
- **Unit Vector Flows**
Joint work with Babak Mirafteb and Pat Morin.
- **Separation Number and Treewidth, Revisited**
Joint work with Babak Mirafteb and Pat Morin. Submitted to Discrete Mathematics in November 2025. [arXiv:2503.17112](#)
- **Connected Dominating Sets in Triangulations**
Joint work with Prosenjit Bose, Vida Dujmovic, Pat Morin, and Saeed Odak. Submitted to Journal of Combinatorial Theory, Series B. [arxiv:2312.03399](#).

Accepted:

- **Linear versus Centred Chromatic Numbers**
joint work with Prosenjit Bose, Vida Dujmovic, Mehrnoosh Javarsineh, and Pat Morin. Journal of Graph Theory. Accepted in March 2024. [arxiv:2205.15096](#).
- **Subquadratic High-Dimensional Hierarchical Clustering**
Joint work with Vincent Cohen-Addad and Amir Abboud. The proceedings of the conference on Neural Information Processing Systems ([NeurIPS 2019](#)).

WORK EXPERIENCE

Since Sep 2021	Teaching Assistant at Carleton University Assisted in the following courses: Algorithms and Data Structures, Computational Aspect of GIS, Discrete Structure II, and Introduction to Theory of Computation.
Apr. 2019 - Aug. 2019	Research Intern at the Sorbonne Laboratory of Computer Science LIP6 Supervised by Prof. Vincent Cohen-Addad. Designed, analyzed, and implemented a Sub-quadratic High Dimensional Hierarchical Clustering Algorithms.
Apr. 2018 - Aug. 2018	Research Intern at LORIA/INRIA Grand Est, Nancy - Alice Team Supervised by Prof. Bruno Levy. Worked on Topological Optimisation algorithm based on Laguerre diagram. A report on this work can be found here .
Jan. 2020 - Jan. 2021	Teaching Assistant at the American University of Beirut Designed and graded assignments. Gave problem solving sessions for the undergraduate's Advanced Algorithms and Data Structures Course. Graded assignments for the graduate Graph Algorithmic Theory Course.

WORKSHOPS AND SUMMER SCHOOLS

Feb 2026	Thirteenth Annual Workshop on Geometry and Graphs At this workshop, a group of researchers will work on open problems in graph theory and geometry.
June 25 - July 25	School on Graph Theory (SGT 2025) Attended courses on Parametrized Algorithms, Polynomials and Graphs, and Quantum Computation Applied to Graphs.
June 24 - July 24	Czech Summer School in Discrete Mathematics Attended courses on Twin-width and Posets.
May 23 - June 3 2022	Berlin Mathematical School - Summer School Attended courses on Topology and Geometry in Discrete Settings.

SCHOLARSHIPS, HONORS, AND AWARDS

2013 - 2017	Awarded full scholarship from The United State Agency for International Development USAID to study BSc. in Computer Science at the American University Of Beirut.
2014 - 2017	Placed four times on the Dean's Honor List during my studies at the American University of Beirut.
2016	Won the first place (team 1729) in ACM - The 8 th Lebanese Collegiate Programming Contest.
2013 - 2014	Awarded two certificates for participating in several volunteering activities from the Center of Civic Engagement and Community Service (CCECS) at the American University of Beirut.

LANGUAGES

English (Fluent), French (Fluent), and Arabic (Native).

COMPUTER SKILLS

Languages: Python, Java, C, C++, Haskell, and JavaScript.

Tools and Libraries: QGIS, Linux, HTML, OpenGL, Git, CMake, NTL, GMP, SFML, and $\text{\LaTeX}$.

INTERESTS AND ACTIVITIES

Hobbies: Climbing and Playing Guitar.